

Challenge (Habanero)

Open Ended Questions (FRQ)

Q9. A water tank can hold 1000 gallons of water. Water is being pumped into the tank at a rate of 5 gallons per minute. If the tank is currently empty, how many minutes will it take to fill the tank? Show your work and explain your answer.

Q10. A squirrel is collecting nuts for the winter. The squirrel's nut collection grows at a rate of 2 nuts per day. If the squirrel currently has 15 nuts, how many days will it take for the squirrel to collect 50 nuts? Show your work and explain your answer.

Chef's Tips

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- Emphasize the importance of units when interpreting slope as rate of change
- Use real-world examples to illustrate the concept of slope and rate of change
- Encourage students to check their work by plugging in values into the equations